STRhub Verified
strhub.app · strhub.app/verified

Technical Reproducibility Validation Report

STRsearch

Version c70179b
Autosomal STR

Verification date	2026-08-14
Repository commit	c70179b3b175adc82a7314409af06900b3861d61
Attestation level	**Runs + Plausible output**
Permalink	https://strhub.app/verified/strsearch-c70179b
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

- Source available
- Installation successful
- End-to-end execution
- Output generated

NOT VERIFIED

- Genotype accuracy
- Concordance
- Forensic validity
- Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**5/5 gates passed. Runs + Plausible output**
ATTESTATION LEVEL	**Runs + Plausible output**
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	STRsearch
Version	c70179b
Repository	https://github.com/AnJingwd/STRsearch
Commit (pinned)	c70179b3b175adc82a7314409af06900b3861d61
Container OS	ubuntu-22.04
Marker panel	Autosomal STR
Verified on	2026-08-14 12:22:21 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/31799926807
Submitted by	A third party (not the tool's maintainer)

This tool was submitted for verification by somebody other than its maintainer. The maintainer took no part in the run and supplied none of what it used: the command, the environment, and any target regions were chosen by the submitter. This certificate records a run; it is not an endorsement by the tool's maintainer, and does not imply their involvement.

3. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · STRsearch c70179b · commit c70179b
$ python3 /opt/STRsearch/pipeline.py \
--type paired \
--num_processors 4 \
--reads_threshold 30 \
--stutter_ratio 0.5 \
from_bam \
--working_path /data/out \
--sample sample \
--sex female \
--bam /data/in/input.bam \
--ref_bed /data/in/regions.bed \
--genotypes /data/out/genotypes.txt \
--multiple_alleles /data/out/multiple_alleles.txt \
--qc_matrix /data/out/qc_matrix.txt
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	strsearch-c70179b.log-external.txt	

5. Out of Scope

This report does not evaluate any of the following:

– Genotype correctness or accuracy

– Concordance against known truth sets

– Sensitivity, specificity, or stutter performance

– Allele calling accuracy or forensic casework suitability

– Regulatory compliance or ISO accreditation

– Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	multiple_alleles.txt
Format	TSV
Sequence records	388
Distinct STR loci	24
Total reads	2,401
Max depth (single locus)	51

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D5S818 · D6S1043 · D7S820 · D8S1179 · FGA · PentaD · PentaE · SE33 · TH01 · TPOX · vWA
Regions BED	Provided by a third party, not the tool's maintainer (covers 24 of 24 supported loci)

8. Verification Matrix

PASS	Reference dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– A regions configuration file, supplied with the submission rather than taken from the repository.
– Test data: no sample from the repository was used, so a public reference sample stood in.
– A container environment, supplied with the submission.

10. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– --ref_bed expects an 11-column STR configuration file (chr/start/end/period/.../flanking sequences), not a plain BED.
– Example coordinates and reference are hg19 (ucsc.hg19.fasta, ref_test.bed); an hg38-aligned BAM needs an hg38 STR config file.
– usearch is required but licensed; the Dockerfile fetches it from a personal license URL, which no other user can reach.
– The from_fastq mode needs a bwa-indexed reference (.amb/.ann/.bwt/.pac/.sa); only hg38.fa and .fai are mounted, so use from_bam.
– The Dockerfile assumes code lives in a ./STRsearch subdirectory, but conf.py, pipeline.py and scripts/ are at the repository root.
– The Dockerfile pins ubuntu:16.04, Miniconda 4.3.31 and old bioconda builds (samtools 1.7, bedtools 2.17.0); these pins no longer resolve.
– --sex is required by from_bam; the example uses male, which affects Y-locus handling.

11. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

12. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

13. Conclusion

◆ **Runs end-to-end**

STRsearch c70179b installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 5 verification gates were passed.

◆ **Produces valid output**

The tool generated a structurally valid TSV output file with genotype calls across 24 target forensic STR loci, with a maximum read depth of 51 at vWA.

◆ **No bundled demo data**

STRsearch does not include its own test or demo dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth

STRsearch c70179b · verified 2026-08-14

LOCUS	READ DEPTH	LOCUS	READ DEPTH
vWA	241	**CSF1PO**	92
D21S11	187	**D6S1043**	92
FGA	164	**D2S441**	88
D19S433	131	**D1S1656**	81
SE33	125	**PentaE**	78
D18S51	123	**D13S317**	66
D16S539	105	**D10S1248**	65
D3S1358	99	**D2S1338**	62
PentaD	99	**D8S1179**	62
D7S820	96	**TPOX**	55
D22S1045	95	**D12S391**	55
D5S818	94	**TH01**	46